NLP and Text Analytics: Final Session

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CBS

Session 1: Introduction

- The Revolution in NLP
- Language is Hard!
  - Ambiguity must be resolved with respect to context – both lexical and structural
  - Dependencies in context can be unbounded
- ChatGPT and other LLM's
  - Appear to have (close to) human-level language ability
  - Exciting research directions – Building applications/ Probing their abilities

Session 2: Basic Text Processing

- Regular expressions
- Text Preprocessing:
  - Tokenization
  - Lowercasing
  - Lemmatization
- NLTK and SpaCy
  - POS tagging
  - NER (Named Entity Recognition)
  - Lemmatization

Session 3: Language Modeling

- Language Modeling
- Ngrams: unigrams, bigrams, trigrams
- Shannon game, Perplexity

Markov Assumption

Andrei Markov

Simplifying assumption:

$$P(\text{the} \mid \text{its water is so transparent that}) \approx P(\text{the} \mid \text{that})$$

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$$P(w_1 w_2 \ldots w_n) \approx \prod_i P(w_i \mid w_{i-k} \ldots w_{i-1})$$

Approximate each component in the product

$$P(w_i \mid w_1 w_2 \ldots w_{i-1}) \approx P(w_i \mid w_{i-k} \ldots w_{i-1})$$

Session 4: Text Classification

- Text Classification
- Naive Bayes Classifiers
- Sentiment and Binary Naive Bayes
- Evaluation: Precision, Recall, F-measure

Session 5: Logistic Regression

- Text Classification with Logistic Regression
- Sigmoid and Probabilities
- Features for Sentiment

Session 6: Sentiment Analysis

- Sentiment Analysis
- Lexicon-based
- Aspect-based
- Emotions

Session 7: Topic Modeling

- Topic Modeling
- LDA
- top2vec
- BERTopic

Session 8: Vector Semantics and Embeddings

- LLMs and AGI (Artificial General Intelligence)
- Vector Semantics – Summarizing word contexts as dense vectors
- word2vec –
  - Powerful representations for many NLP tasks: translation, sentiment, QA, . . .
  - Learned with skip-gram word prediction method
  - Black box: what is really being learned?
  - Abstract relations like: male-female, or capital-city-of, or even comparative/superlative
  - Static embeddings: word embedding doesn't change with context

Session 9: Simple Neural Networks

- Why Language is Hard: Lexical and Structural Ambiguity Relies on Context
- Neural Networks
  - Units in Neural Networks
  - Non-linear activation functions
  - XOR Problem
  - Feedforward Neural Nets for Language Classification
  - Embeddings as Input Features

Session 10: Large Language Models and the Transformer

- Transformers use attention mechanism to allow word representations to be enriched by context
- BERT is encoder model suited for classification tasks like sentiment analysis or ticket routing
- GPT is decoder model suited for instruction or chat
- Prompt engineering – powerful techniques for interacting with LLMs

Session 11: Post-training – Question Answering and Chatbots

- Post-training
  - instruction following, can improve reasoning chains for any task
- Working with LLMs
  - Prompt engineering
  - local LLMs
  - LLMs and API

General Issues Raised by Current Models

- AI Alignment
- Explainable/Interpretable AI
- Science vs. Engineering in AI
- AGI: where is AI heading?